

EasyPaint

“An easy to use art program for beginners, the elderly, and the disabled.”

by Bradrick Rush

Problem Statement

- Current art tools of today:
 - Too complex for beginners.
 - Lack of accessibility tools for the elderly or disabled.
 - Expensive and/or relies on subscription plans.



Photoshop

Save 40% on Photoshop for the first six months. Ends May 12. [See terms.](#)

~~US\$22.99/mo~~ **US\$13.79/mo**

Annual, billed monthly

Best for	Individuals
7-day free trial 	✓



The Solution: EasyPaint

- A simplified user interface.
 - A toolbox with only the essentials, but can be customized.
- Support for tablets and other touchscreen devices.
 - Gestures.
 - Undo and redo.
 - Natural drawing.
- Accessibility features:
 - Voice commands.
 - UI adjustments (contrast and font size).
 - Support for alternate inputs.



Target Users

- Beginners
- Elderly users
- Disabled users
- Affordable
 - 30-day free trial.
 - Low cost one-time purchase; no subscription.
- Devices:
 - Tablets
 - Laptops
 - Desktops

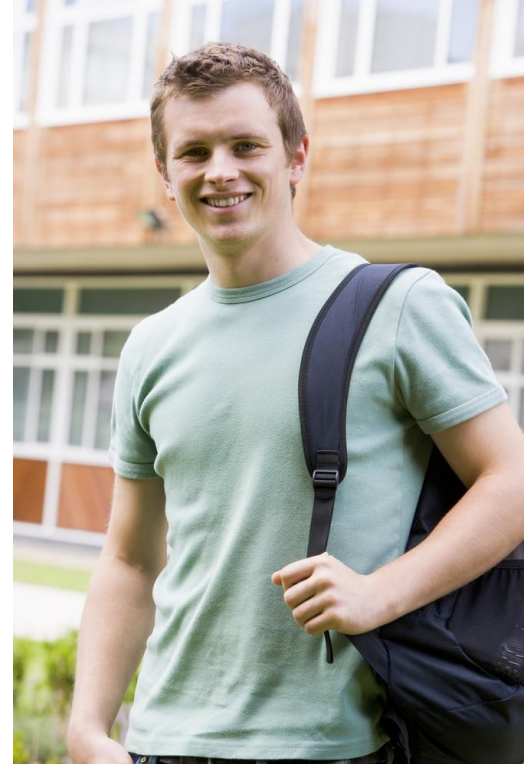
Personas

- Name: Cassandra Wilmar
- Occupation: Retired School Teacher
- Goal: Wants an easy to use art program with minimal learning curves.
 - Loves EasyPaint for voice commands and large readable buttons.



Personas

- Name: David Balmier
- Occupation: College Art Student
- Goal: Wants an affordable art program that is very responsive and simple to use.
 - Loves EasyPaint for keyboard commands and tablet gestures.



Personas

- Name: Adrian Jones
- Occupation: Freelance Designer
- Goal: Wants an art program that can accommodate his motor disability.
 - Loves EasyPaint for its support for alternate input and voice-activated commands.



Empathy Map Insights

- Consumer avoids complexity and inconsistency; discourages use.
- Consumer praises simplicity and straightforwardness; encourages use.

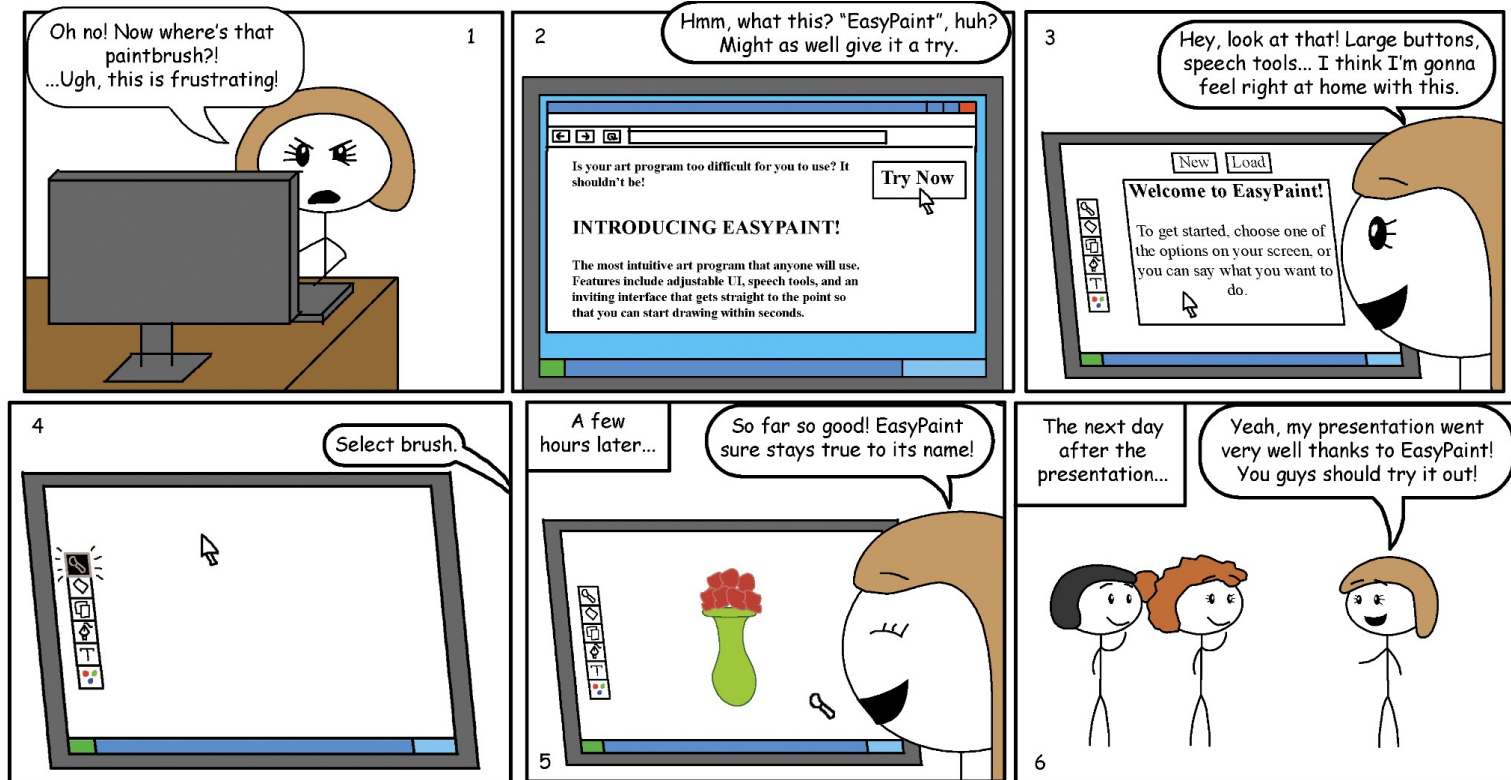
Empathy Map

Says	Thinks
● “I don’t know much about computers.” ● “This seems very confusing.”	● “I want to express my creativity again.” ● “Why can’t this be simpler?”
	
Does	Feels
● Takes breaks when frustrated. ● Avoids complex menus.	● Intimidated. ● Optimistic. ● Proud when something works.

Supporting Data

- Visual impairments:
 - Smaller or lower contrast interfaces difficult to decipher.
 - Affected users:
 - Disabled.
 - Elderly: Around 60% of Americans aged 65 or older.
- Cognitive load:
 - Complex user interfaces encourage it.
 - Greatly affects the disabled and the elderly.
- Motor impairments:
 - Upper limb impairments prevent use of standard tools.
 - Alternative inputs are required.

Storyboard



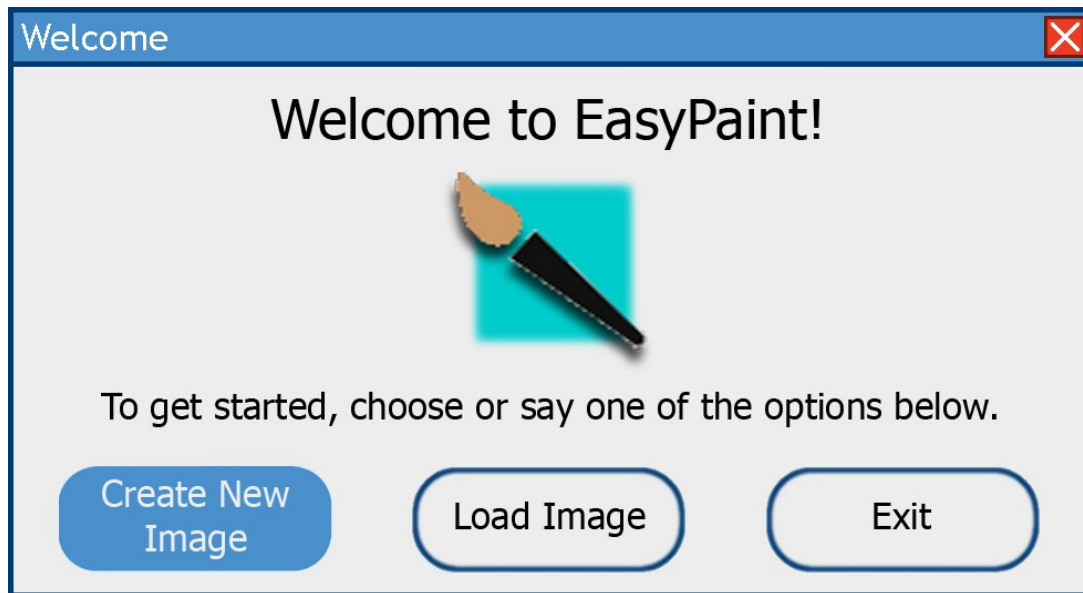
EasyPaint Simulation (Begin)

Click here to
begin.



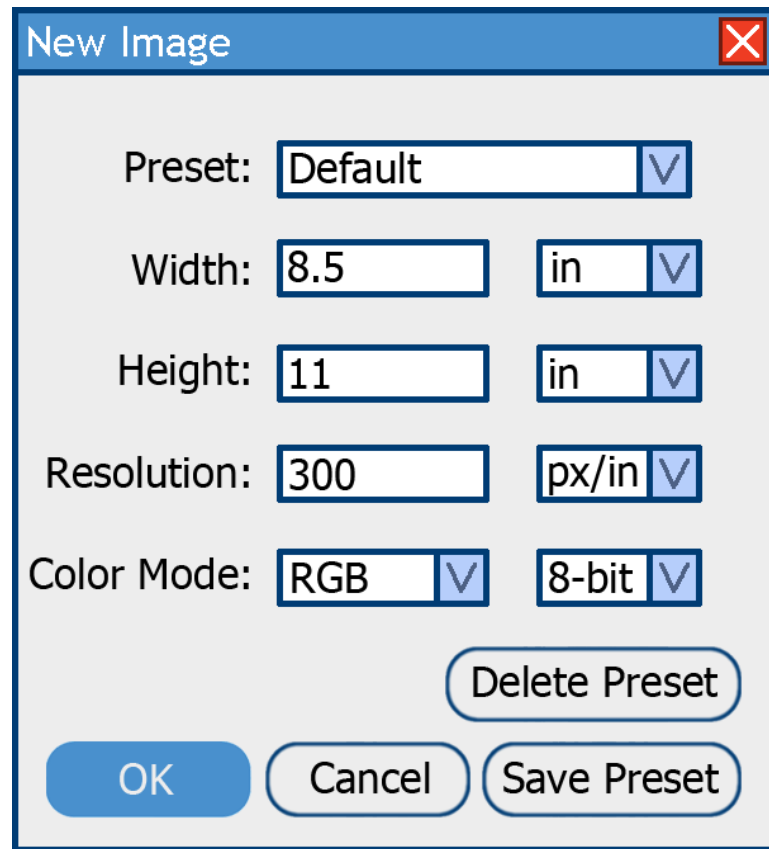
EasyPaint Simulation

- Welcome Screen
 - Create New Image
 - Load Image
 - Can click or say
- Let's Create New Image!



EasyPaint Simulation

- Create New Image
 - Options:
 - Width
 - Height
 - Resolution
 - Color Mode
 - Presets:
 - Quick settings for each image
 - Customizable
- We're ready to go!



A screenshot of a 'New Image' dialog box. The title bar is blue with the text 'New Image' and a red close button. The dialog has a light gray background. It contains several settings: 'Preset' is a dropdown menu showing 'Default'; 'Width' is a text box with '8.5' and a unit dropdown showing 'in'; 'Height' is a text box with '11' and a unit dropdown showing 'in'; 'Resolution' is a text box with '300' and a unit dropdown showing 'px/in'; 'Color Mode' is a dropdown menu showing 'RGB' and a bitness dropdown showing '8-bit'. At the bottom right is a 'Delete Preset' button. At the bottom are three buttons: 'OK' (blue), 'Cancel' (white with blue border), and 'Save Preset' (white with blue border).

New Image

Preset: Default

Width: 8.5 in

Height: 11 in

Resolution: 300 px/in

Color Mode: RGB 8-bit

Delete Preset

OK Cancel Save Preset

EasyPaint Simulation

Main Interface

Accessibility Options

Big arrow or swiping here shows/hides left buttons.

Toolbox:

- Brush
- Eraser
- Layers
- Paths
- Text
- Zoom

Color Picker

Customize Toolbox

Big arrow or swiping here shows/hides top buttons.

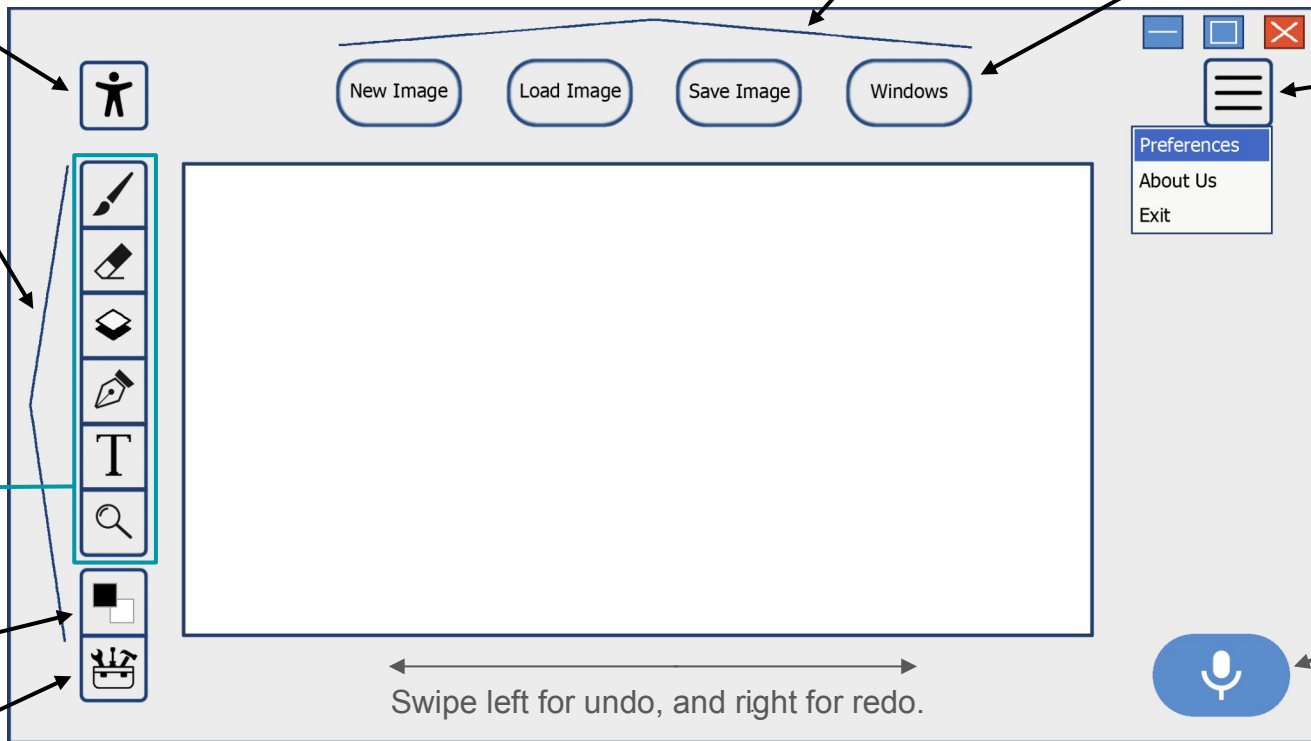
Manages multiple open images.

Menu

Preferences
About Us
Exit

Voice commands
(Activated by click or Hotkey)

Swipe left for undo, and right for redo.



EasyPaint Simulation

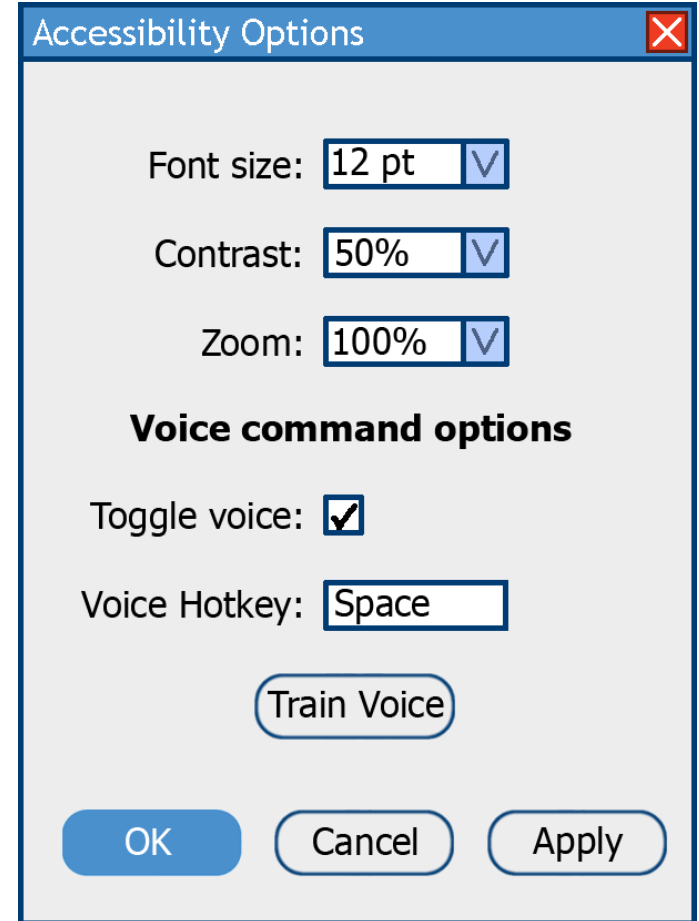
- Accessibility Options

- Options:

- Font size
 - Contrast
 - Zoom

- Voice command options:

- Toggle
 - Hotkey
 - Train Voice



The screenshot shows a dialog box titled "Accessibility Options" with a red close button in the top right corner. The dialog has a light gray background and a blue border. It contains three settings: "Font size" set to "12 pt", "Contrast" set to "50%", and "Zoom" set to "100%". Each setting has a dropdown arrow to its right. Below these is a section titled "Voice command options" in bold. Under this section, there is a "Toggle voice" option with a checked checkbox, and a "Voice Hotkey" option with a text field containing "Space". At the bottom of the dialog, there are three buttons: "OK", "Cancel", and "Apply".

Accessibility Options

Font size: 12 pt ▼

Contrast: 50% ▼

Zoom: 100% ▼

Voice command options

Toggle voice: ☒

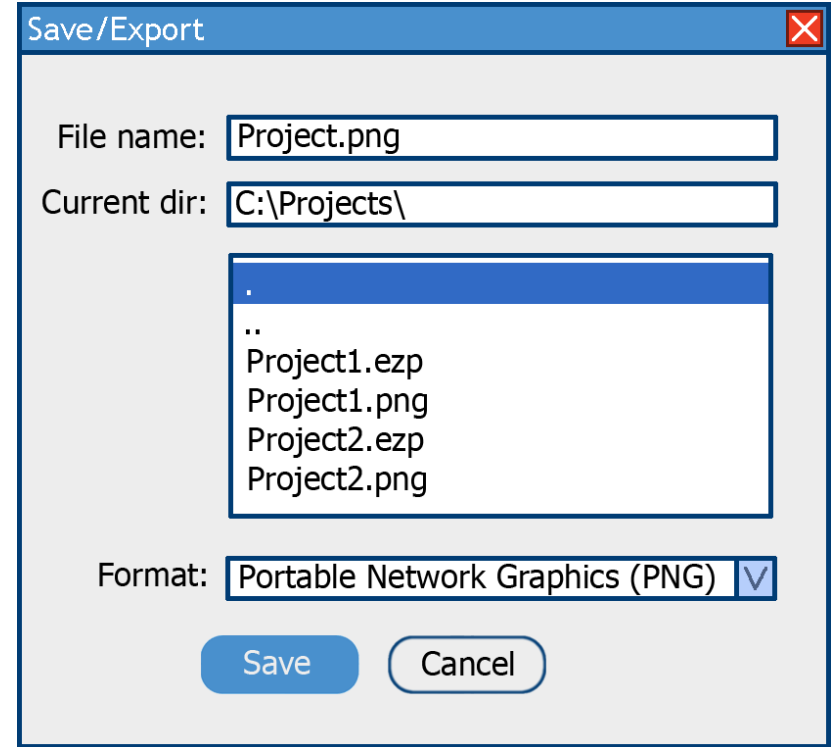
Voice Hotkey: Space

Train Voice

OK Cancel Apply

EasyPaint Simulation (End)

- Save/Export
 - File name
 - Current directory
 - Directory listing
 - File formats:
 - Common (PNG, JPEG, etc.)
 - Unique: EZP for EasyPaint Project
 - Saves Layers, Paths, etc.



Conclusion

- EasyPaint provides an easy way for users to express themselves digitally.
 - Accommodative for the elderly and the disabled.
 - No complexity, no inconsistencies, and is a viable low-cost solution in the long run.



References

UX design for Seniors: Examples and tips. (2025b, March 18). <https://www.eleken.co/blog-posts/examples-of-ux-design-for-seniors>

Zhou, C., Yuan, F., Huang, T., Zhang, Y., & Kaner, J. (2022). The Impact of Interface Design Element Features on Task Performance in Older Adults: Evidence from Eye-Tracking and EEG Signals. *International Journal of Environmental Research and Public Health*, 19(15), 9251. <https://doi.org/10.3390/ijerph19159251>

Cossovich, R., Wu, S., & Girouard, A. (2024). I tried everything. Nothing works: Challenges and Creative Processes from Digital Artists with Upper Limb Motor Impairments. *ACM Digital Library*, 4, 1–17. <https://doi.org/10.1145/3663548.3675654>

Images Used

- https://miro.medium.com/v2/resize:fit:1400/1*oJcYafbNTeDMbVICne169A.png
- <https://images.vexels.com/media/users/3/135157/isolated/lists/a9e180fec4b12df72db594a79b8f65e9-microphone-on-button-flat-icon.png>
- <https://creativevip.net/resource/21-photoshop-toolbar-icons/>
- <https://static-00.iconduck.com/assets.00/accessibility-icon-1665x2048-m9ypfxgq.png>
- https://static.vecteezy.com/system/resources/previews/020/975/526/non_2x/window-10-go-window-10-icon-transparent-free-png.png
- https://static.vecteezy.com/system/resources/previews/039/302/860/non_2x/blank-screen-tablet-frame-layout-of-a-universal-set-of-devices-ui-tablet-mockup-for-ux-for-infographics-or-presentations-vector.jpg